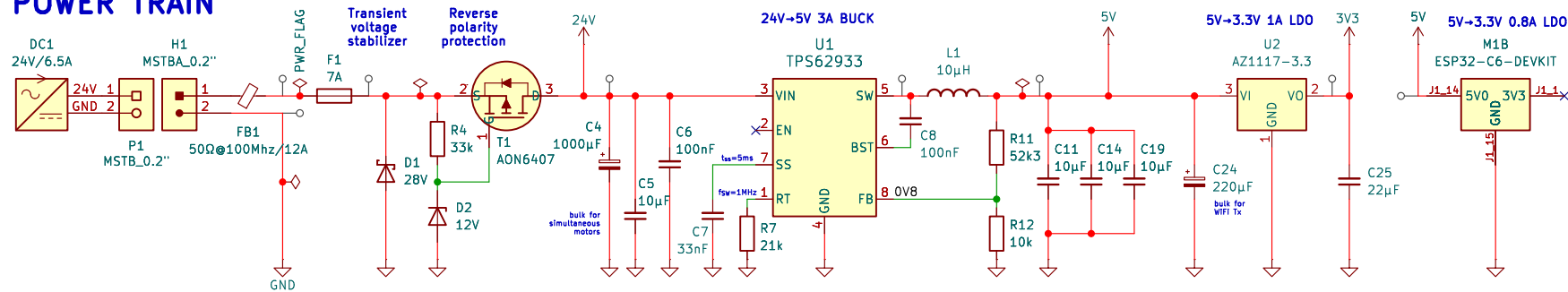
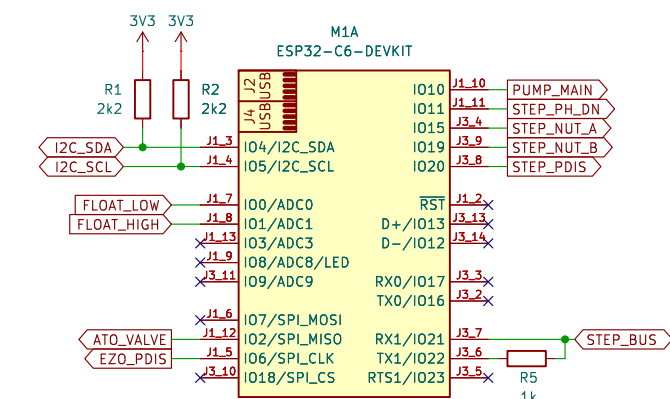


POWER TRAIN

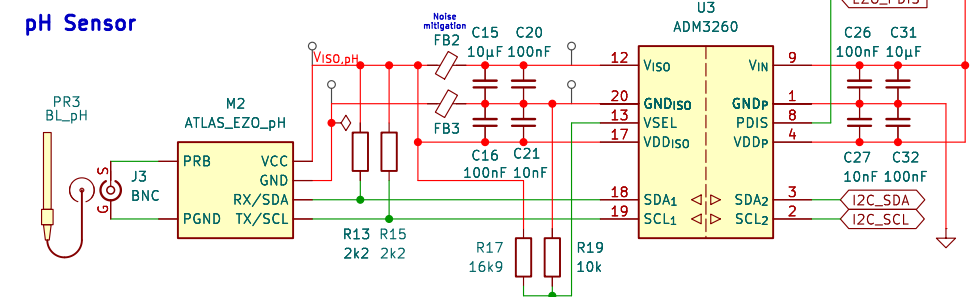


SoC

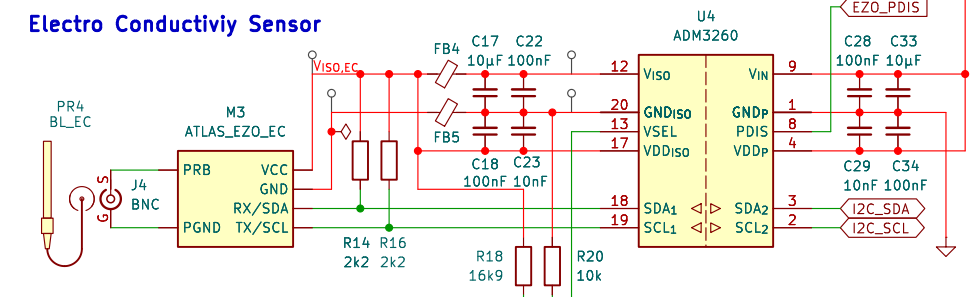


I2C SENSORS

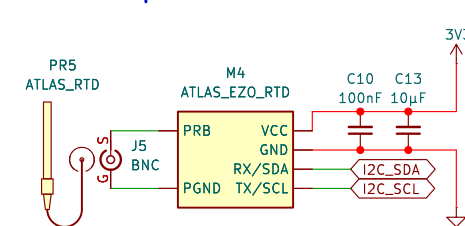
pH Sensor



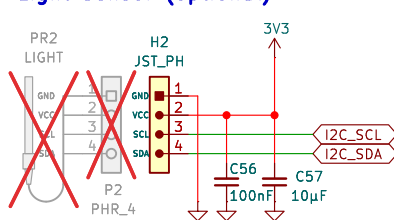
Electro Conductivity Sensor



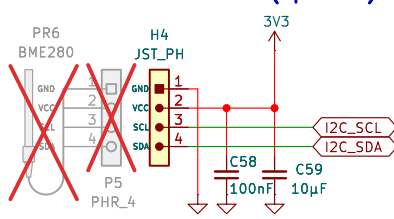
Water Temperature Sensor



Light Sensor (optional)

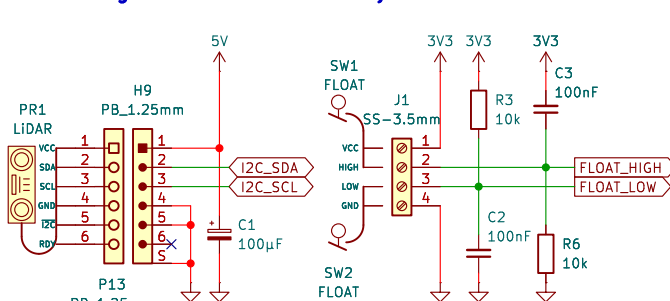


Environmental Sensor (optional)

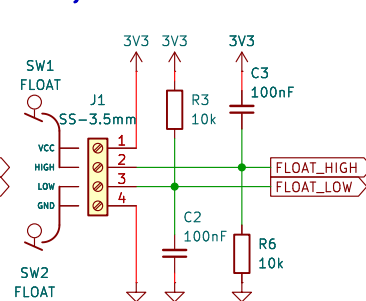


WATER LEVEL SENSOR/SWITCHES

Water Height Sensor

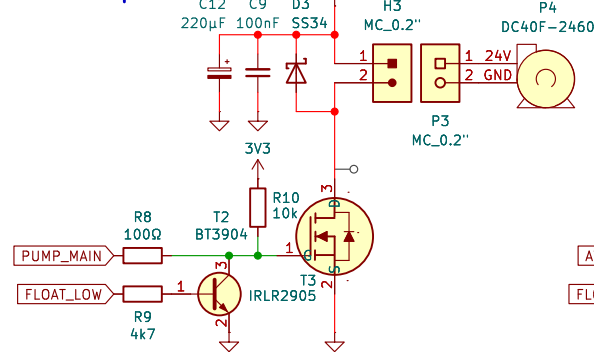


Safety Switches

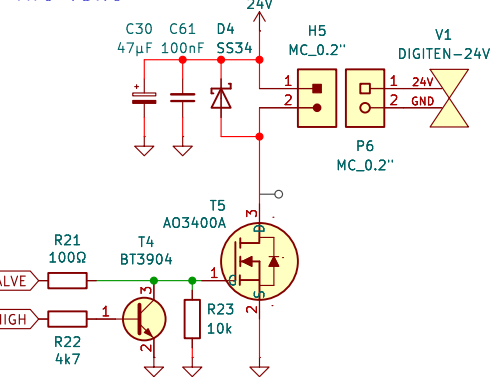


MAIN PUMP & VALVE DRIVERS

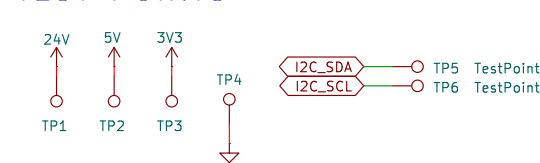
Main Pump



ATO Valve

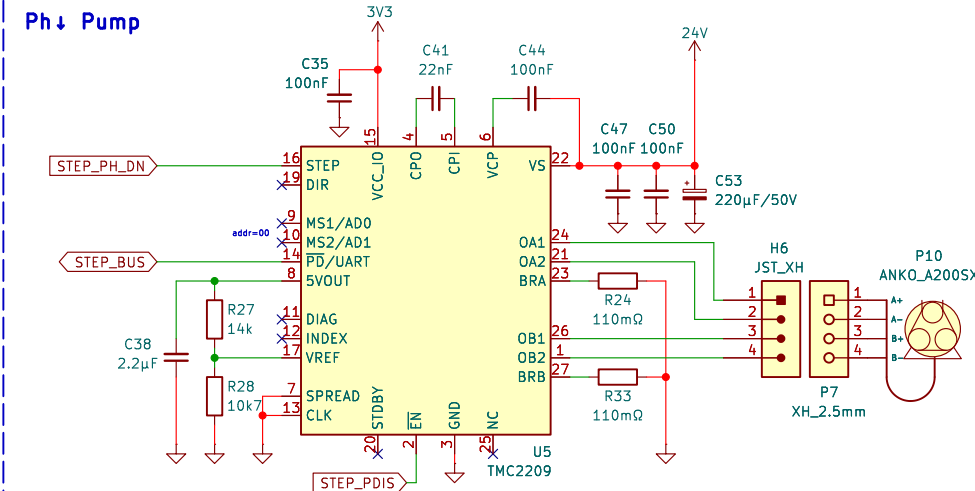


TEST POINTS

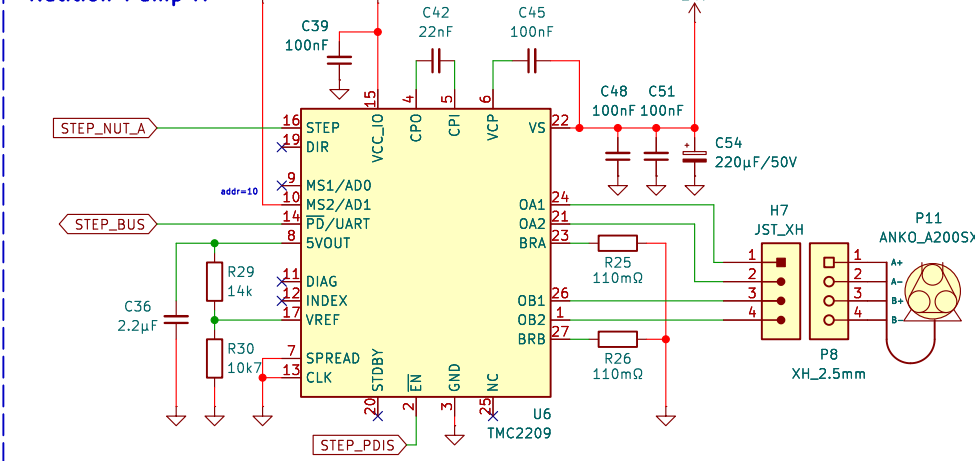


PERISTALTIC PUMP DRIVERS

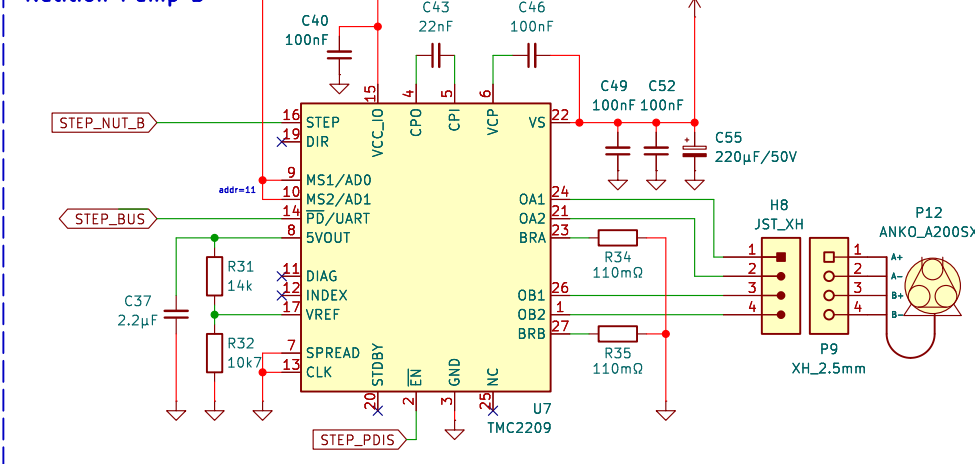
Ph Pump



Nutrition Pump A



Nutrition Pump B



Autonomous, data-rich hydroponics irrigation automation with precision control and remote monitoring. ESP32-C6 based with full sensor suite, dosing pump control, and automatic top-off (ATO) capability. Designed for Home Assistant or Matter over Thread/WiFi.

<https://github.com/cvnon/OPNhydry>

Sheet: /

File: OPNhydry_r2.kicad_sch

Title: OPNhydry

Size: A3 Date: 2026-03-03

Rev: r2

KiCad E.D.A. 9.0.7

Id: 1/1